Industrial Internet of Things Solution for Real-Time Monitoring of the Additive Manufacturing Process

Mahmoud Salama[1], Ahmed Elkaseer[2, 3], Mohamed Saied[1], Hazem Ali[4], Steffen Scholz[2]

[1] Faculty of Engineering, Ain Shams University, Cairo, Egypt
[2] Karlsruhe Institute of Technology, Karlsruhe, Germany
[3] Faculty of Engineering, Port Said University, Port Said, Egypt
[4] European X-ray Free-electron Laser Facility GmbH, Schenefeld, Germany
ahmed.elkaseer@kit.edu

Abstract. Modern cyber manufacturing has been introduced into a broad range of manufacturing processes to ease their digital reconfigurability and enhance flexibility while retaining a high throughput of quality products. Such a system provides real-time data acquisition, enabling monitoring of the actual condition of the manufacturing process. The Industrial Internet of Things (IIoT) facilitates such real-time monitoring and optimization of the fabricating system, which reduces time necessary for maintenance, with the possibility of almost instantaneously taking any necessary corrective measures with respect to either human to the machine/process from learned algorithms. In this research, an original IIoT approach has been proposed to monitor the process conditions, including nozzle temperature and filament breakage/runout, of the additive manufacturing process. In particular, concurrent multi-task IIoT communication was developed for a network of five nodes. This was implemented to ensure real time monitoring of the manufacturing process via multi sensors and empowered by an embedded software design. The proposed embedded software architecture offers a reliable solution to eliminate communication latency and provides real-time response to acquired information. It is worth emphasizing that the embedded software was designed so that it optimally exploits the very great potential of the hardware resources, with the ability to detect run-time issues in the nodes' performance and re-try to address such issues to maintain a high capability networking communication. The designed architecture also offers auto-scaling throughput of the data transferred to the cloud to minimize the bandwidth.

Keywords: IIoT, M2M, Industrial Informatics, Industry 4.0, AM.

1 Introduction

The advent and subsequent explosive growth of advanced information- and communication-driven technologies have promoted digitalization in manufacturing industries [1], aiming at realizing the potential of cross-linking intelligent manufacturing operations with automated near-real-time data acquisition and simulation technologies [2]. This enables the so-called cyber/physical modelling of manufacturing processes, which

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is considered the key enabling technology for the transformation to the next industrial revolution, Industry 4.0 [3]. The introduction of such smart cyber/physical modelling into manufacturing technologies includes integrating physical systems with cloud/edge computing, machine/deep learning algorithms, internet of things (IoT) and process simulation techniques, where cyber and physical worlds liaise profitably [4]. This approach offers fast and accurate prediction of manufacturing process outcomes, provided there is a comprehensive and seamless linkage between the physical and digital worlds. In particular, there is the possibility of instantaneous acquisition of physical problems with the almost concurrent production of necessary corrective actions, which it is expected will ultimately help optimize the performance of entire manufacturing systems [12]. However, among the aforementioned fragments of Industry 4.0, IoT is a central part that facilitates the interconnection between the intelligent manufacturing assets in a timely and effective manner. In particular, IoT, particularly Industrial IoT (IIoT) allows the manufacturing system, sub-systems, raw materials, and products to be in contentious communication sessions from the supply-chain to the final manufacturing stage. It allows for flexible and customizable production, which increases the manufacturing productivity, and flexibility, which positively reflects on the economic aspects [5].

The IIoT paradigm is presently based on two technologies: radio frequency identification (RFID) and wireless sensor networks (WSN) [6]. Whereas RFID is a microchip-based technology that interfaces with an antenna and presents identification through radio waves to define the objects, WSN is a group of interconnected intelligent sensors, which monitor a certain object for a predefined purpose.

The implementation of IIoT technology entails a series of design layers. The first is the data acquisition layer that consists of groups of intelligent sensors' nodes responsible for sensing the signals and processing these signals to collect the required data. These sensors are designed to perform the data acquisition techniques in different domains such as invasive or noninvasive techniques. The second IIoT layer is the network layer, which allows the nods to interact/communicate with each other to exchange the acquired information. Furthermore, the network layer works to concatenate data from other networks and to establish connections between all heterogeneous systems through the internet. Thirdly, the service layer is defined as the middleware IoT technology that integrates the application and the service [7], to provide application requirements such as APIs, and to implement the control process for information storage and data management.

The design of IoT solution is facing many challenges related to the feasibility of the proposed solution to facilitate effective and seamless mass communication among the manufacturing assets, to minimize the independency of nodes to each other, which would allow for a stable solution under different working conditions [8], and to manage the throughput capacity of the transferred data.

In this context, the aim of this paper is to develop an original Industrial Internet of Things Solution to model the physical-cyber world of the additive manufacturing (AM) process to real-time monitoring of AM fabricating conditions. For demonstration purposes, the conditions of the AM process to be monitored are the temperatures of the

nozzle and filament breakage. Besides, human-machine interactions have been developed based on local and remote techniques that offer monitoring of the additive manufacturing processes directly by the operator.

2 Proposed IIoT Solution Architecture

2.1 Physical Architecture

The proposed IIoT network structure presents a real-time solution for monitoring and interfacing with the additive manufacturing process. The design is empowered by a convenient communication algorithm based on a wireless approach. The network structure includes sensing nodes, a HMI node, a server mirror node and a control unit node.

Sensing Nodes (two sensors): responsible for sensing, acquiring, and processing the sensing parameters of the fabrication process.

HMI Node: produces a user-friendly interface and enables the user to investigate parameters of the fabrication process to tune the actions of the fabricating conditions that might be required [9].

Server Mirror Node: facilitates acquisition of all sensed information and sends it to the cloud; it also operates as a backup for the cloud computing in case of issues with the internet connection.

Control ~~unit~~ Unit (IoT Broker): is the main handler of the entire modules, and manages mainly the information exchanges between the modules, checks the robustness of modules performance, investigates any run time issues and tries to solve them.

2.2 Communication Layer

The objective of the communication module is to manage the exchange of information through the nodes channel, and between the node network and the application network. The communication software algorithm of the broker is designed to manage multiple nodes in the network. In addition, the proposed software works to collect, split, and send the information from/to the other solution nodes. The communication between the broker and the other nodes is carried out via a convenient software algorithm to enable high-speed communication response.

2.3 Application Layer

The final layer of IIoT design is the application tier. The fabrication process state has been monetarized based on real-time HMI (Human-Machine Interaction) using an Android-based application. The Android-based application monitors and manages the AM process by using real-time commination with the broker. Also, cloud computing has been used in this design to store, manage and present the in-formation of the AM. Cloud computing has been established based on the connection with the server mirror node, which accesses the internet, formulizes the data and then sends it to the cloud. However,

if there is problem with the internet connection, the node will automatically save the information on the local storage and send it back when the connection is reinstated.

3 Implementation of IIoT Solution

The designed IIoT solution is based on five nodes: two sensor nodes, a server mirror node, an IoT Broker node and a HMI node. The implementation of the proposed idea is as follows: an Android-based application is utilized for the HMI, and other nodes have been designed and developed based on embedded system architecture. The wireless communication between the nodes has been carried out using IEEE 802.11 communication protocols. IoT MathWorks application (ThingSpeak [10]) has been utilized as Cloud computing for the solution, as shown in figure 1.

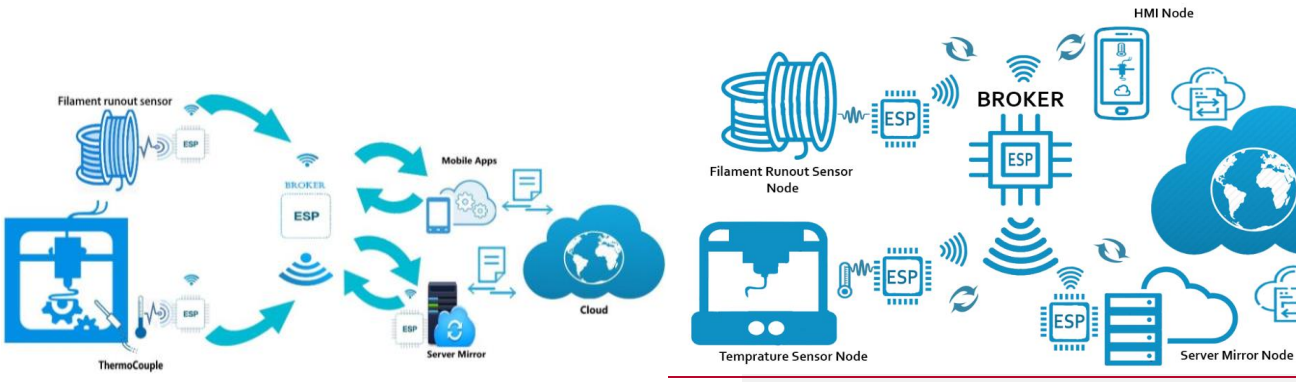
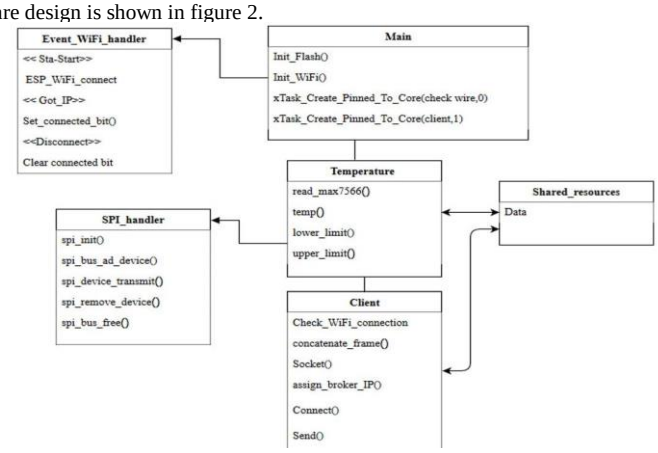


Fig. 1. Proposed Design of the IIoT Solution

The embedded software has been designed based on RTOS (Real-Time Operating System), due to its ability to handle multi-tasks simultaneously. RTOS categorizes the functions of the software into tasks and implements scheduling during the execution process based on demand [11]. Furthermore, RTOS facilitates the embedded software to exploit the high potential of the hardware resources. Thus, it makes all resources active concurrently which entirely eliminates the latency in the system. FreeRTOS and C Programming Language have been utilized to develop the embedded software of the nodes based on the ESP32 module, which includes Dual Core xTensa Microcontroller with a large number of peripherals, Wi-Fi and BLE (Bluetooth Low Energy).

The first sensor node has been designed to measure the temperature of the additive manufacturing nozzle. The design utilized an ESP32 module and a thermocouple K type temperature sensor. This thermocouple sensor measures the temperature up to +1024 °C. The interface between the ESP32 and the Thermocouple is performed by SPI (serial prefer interface) communication protocols through a MAX6675 Module.

MAX6675 Module digitizes the analog signal from a type-K thermocouple to 12-bit resolution via SPI interface with accuracy 0.25 °C. The embedded software design has used the ESP32 interface with Max6675 thermocouple through an SPI bus. Besides, the design allows for a communication session with the broker using the IEEE 802.11 protocol. The multiprocessing is utilized to perform two functions to avoid task blocking and communication latency since communication has been implemented using socket blocking API's. The first task of the multiprocessing is to interface with Max6675 via the SPI which is pinned to core 0 of the ESP32. The second task, a client task, is to connect to the broker and its pinned to core 1 of ESP32. The described embedded software design is shown in figure 2.



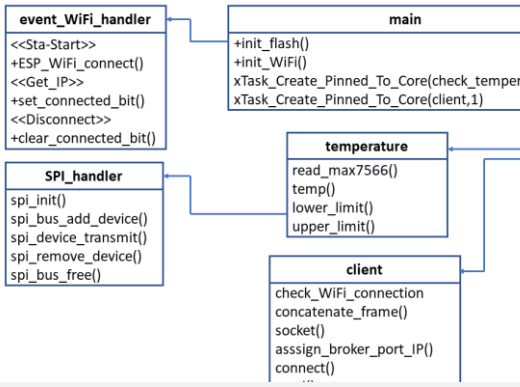


Fig. 2. Embedded software architecture of the temperature sensor node

The second sensor node is the filament breakage detection sensor node. This sensor has been devoted to detect the breakage of the 3D printer filament. In this proposed design, the sensor has been replaced with a switch to perform the same function as the sensor prototype. The embedded software is interfaced to the ESP32 with filament breakage sensor via an I/O port. In addition, the communication between the node and the broker is performed via the IEEE 802.11 protocol. The multiprocessing is utilized to perform these node functions. Thus, core 0 of the ESP32 will perform the breakage sensor function and core 1 of the ESP32 will perform the client task to communicate with the broker, as presented in embedded software design shown in figure 3.

The software architecture for the broker, the main node, has been designed to perform four tasks concurrently, namely three server tasks and one client task. The software architecture is designed to handle, collect and split information via socket communication with all nodes (temperature sensor node, filament breakage sensor node, server mirror node, android-based application). Multiprocessing and multithreading has been utilized in the embedded software design for the broker. Every two tasks are pinned to one of the cores of the ESP32. In addition, every core has multi-tasks as threads. The broker software design diagram is presented in Figure 4.

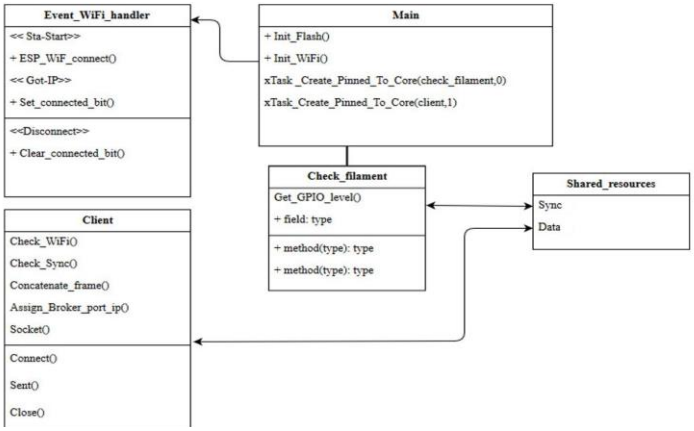


Fig. 3. Software architecture of filament breakage/runout detection sensor node

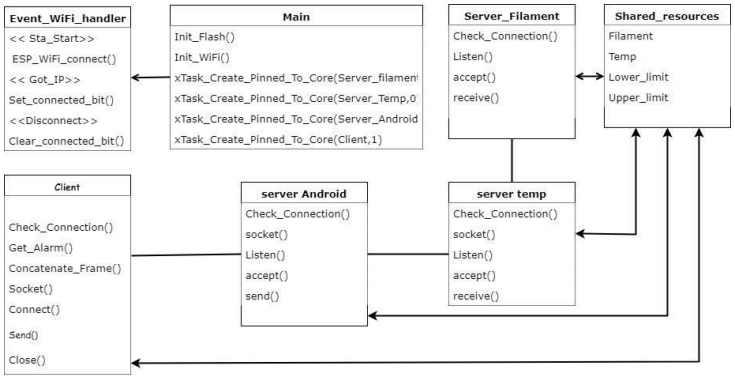


Fig. 4. Software architecture of broker node

The server mirror node has been designed by ESP32 with a SD card interface via a SPI bus. The power technique is battery-supported as the brokers' design is to acquire, save and upload the data whether the electricity on the machine is on or off . The embedded software design of the server mirror node checks the DHCP (Dynamic Host Configuration Protocol) connection of the internet. As long as there is a functioning internet connection, all fabrication process data will be sent to the cloud otherwise during a disconnection situation, the data will be stored locally on the SD card as an offline log.

When the connection is reestablished, the server mirror node will upload the offline log data to the cloud. Figure 5 presents the finite state machine of the server mirror task.

The Android-based application is designed to set the limits of the nozzle temperature and to detect the filament breakage, as shown in figure 6.

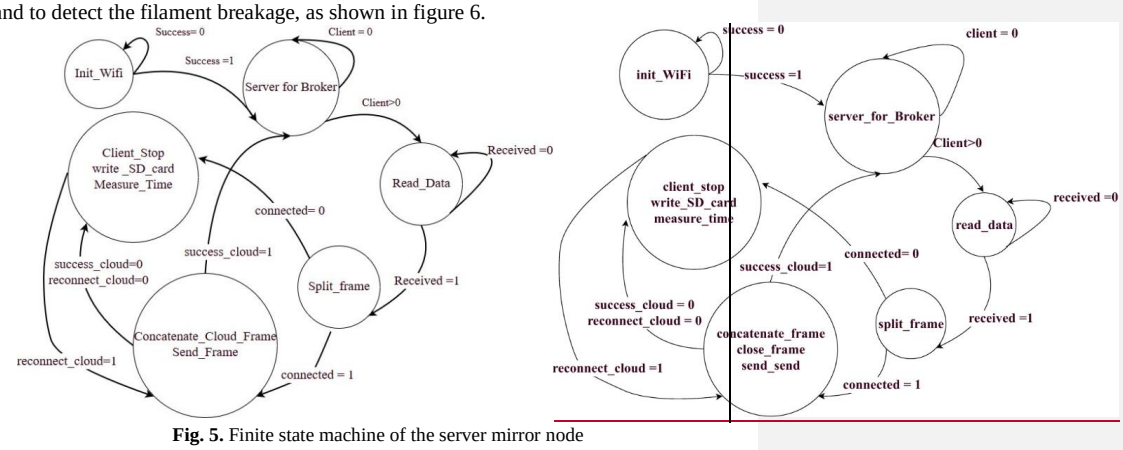


Fig. 5. Finite state machine of the server mirror node

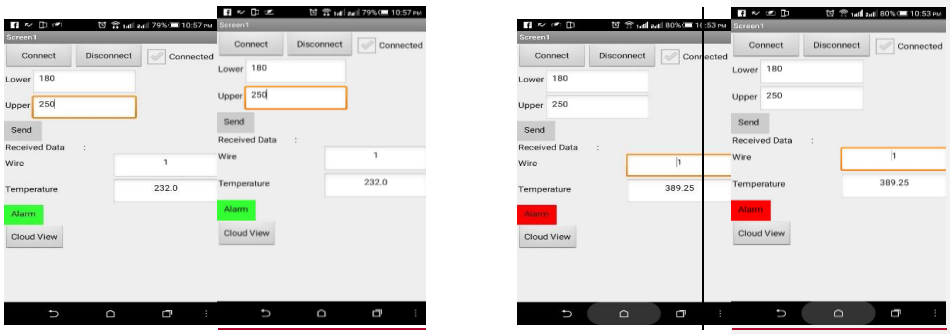


Fig. 6. GUI of Android-based application

The proposed dynamic structure of the nodes' embedded software presents as an activity diagram which shown in figure 7. The working scenario is performed as follows: the broker is defined as the main controller; it is concatenating the sensing data from the sensor nodes into the frame and sends it to the cloud via the server mirror node. In parallel, the data is also sent the android-based application to acquire thermal limit values. The inspector (user) will set the level of temperature that is suitable for the selected AM process via the Android-based application. The broker checks the sensed temperature and compares it to the original setting to identify potential deviations. In case of significant deviations, the broker will add an alarm signal to the concatenated frame of

data. Likewise, an alarm signal will be added if the filament breakage sensor reports a signal read. In case of disconnection to the internet, the uploading of the collected data is impaired and the node will decide upon the storage place for the data (specific local memory storage) and send the data to the cloud once the connection is reestablished.

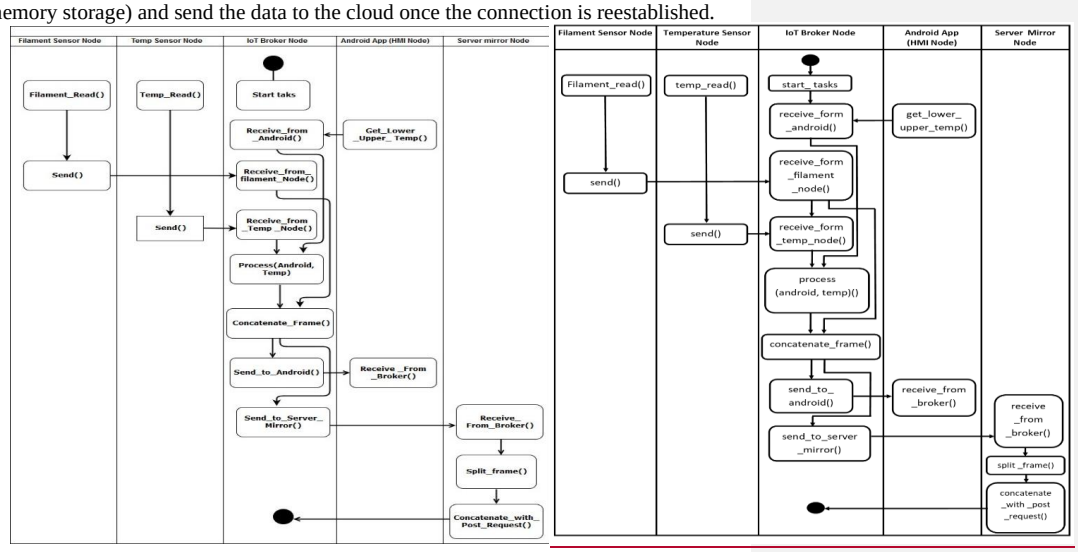


Fig. 7. Activity diagram of the whole solution

4 Results and Discussion

Figure 8(a-d) presents the data visualization on the cloud, where the results are plotted verses time with a time interval of 2 second to check the nozzle temperature (Figure 8a) and the condition of the filament (Figure 8b), where the alarm chart (Figure 8c) presents the response in case of issues detected in the monitored processing conditions. If there is a breakage in the filament or the temperature reading is not in the range displayed on the android-based application, this will be considered an issue. For instance, in our example the temperature limits were set between 180-250 ºC. Figure 8a shows that the temperature varied between 115 and 128 º, which is below the acceptable range. This was recognized as an issue therefore triggering the alarm as displayed in Figure 8c. Wire breakage was also detected (Figure 8b), which indicates the feasibility of the monitoring system to accurately detect the conditions of the fabricating process. Figure 8d shows the data transfer throughput clearly presents the ability of the system to implement auto-scaling based on the uploaded information, which helps to optimize the bandwidth.

To demonstrate the ability of the software solution to handle unexpected behavior that could result in a communication issue between the nodes, the Android application

was disabled on purpose as shown in Figure 9. The results verify the stability of the system, as it continuously performed its functions and once the Android-based application was activated again, a new communication session was established to exchange information with the Android-based application nodes.

Fig. 8. Data visualization on the cloud

```
Received Data  From Wire=W,1  size: 3

Received Data  From Temperature=T,208.75  size: 8
Connected
□[0;32mI (162402) Main_Controller: ... connected□[0m
Error Send
□[0;31mE (162402) Main_Controller: ... socket send failed□[0m

Received Data  From Wire=W,1  size: 3

Received Data  From Temperature=T,207.0  size: 7
lower limit180,upper limit250in client task temeprature=207
Send to Mirror=M,1,207.0,0,22
□[0;32mI (164502) Main_Controller: ... allocated socket
□[0m
```

```
Received Data  From Wire=W,1  size: 3

Received Data  From Temperature=T,208.75  size: 8
Connected
□[0;32mI (162402) Main_Controller: ... connected□[0m
Error Send
□[0;31mE (162402) Main_Controller: ... socket send failed□[0m

Received Data  From Wire=W,1  size: 3

Received Data  From Temperature=T,207.0  size: 7
lower limit180,upper limit250in client task temeprature=207
Send to Mirror=M,1,207.0,0,22
□[0;32mI (164502) Main_Controller: ... allocated socket
□[0m
```

Fig. 9. Self-correction error handling by serial monitoring of the broker

As previously discussed, the proposed activity scenario between the Android-based application, the other nodes and the IoT broker was detected via the serial monitoring of the broker, as shown in Figure 8. The execution time of the entire working solution from the sensing phase until the data delivery to the cloud was 2000 ms, 700 ms for the server mirror task while sending the information to the cloud and other communication between the nodes, data acquisition and data processing took 1300 ms.

5 Conclusion

In this research, a concurrent multi-task IIoT communication solution was developed aiming at real time monitoring of various additive manufacturing processing conditions, in our case the nozzle temperature and filament breakage. The network structure includes several sensing nodes, a HMI node, a server mirror node and a control unit node with the design of the IIoT network being enriched by an auto-scaling throughput of the data transferred to the cloud thereby minimizing the bandwidth. However a convenient communication algorithm based on a wireless approach is utilized to eliminate communication latency and enables the system to handle real time communication issues.

Acknowledgment

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